Harshita Adlakha

Making AI safe enough to fly.

ECE undergraduate researching verification of AI in safety-critical aerospace systems.

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EDUCATION

Bangalore Institute of Technology

B.Tech, Electronics & Communication Engineering
VTU · 2024–2028 · GPA 7.9

PUBLICATIONS

An Architectural Lifecycle Framework for the Verification and Validation of AI-Enabled Aerospace Systems
AIAA SciTech 2027 Forum · accepted

A Sociotechnical Framework for Verifying Non-Deterministic AI in Safety-Critical Aviation Electronics
AIAA SciTech 2027 Forum · accepted

Socio-Technical Complexity Management in Next-Generation Avionics Systems
IAF 77th International Astronautical Congress · Oct 2026

CERTIFICATIONS

Google Cloud Certified: Generative AI Leader
IBM Qiskit Advocate
Airtable: Agentic Systems Design
IBM SkillsBuild: Software Engineering
Google Cloud Professional Cloud Architect (in progress)

SKILLS

Python · C/C++ · Verilog · MATLAB · SQL
PyTorch · XGBoost · SHAP · LLMs · RAG
GCP · Vertex AI · Cloud Run · ADK
MBSE · systems architecture · Git

EXPERIENCE

Research Associate, Blue Marble Space Young Scientist Program

Jun–Aug 2026 · under Dr. Jen Blank

AI-assisted design of human habitats on the Moon and Mars: systems-level concepts using in-situ resource utilisation. Invited to present at BlueSciCon Forum 2026.

PROJECTS

FDIR Copilot · satellite fault diagnosis agent

Accel AI Innovate Paris hackathon · Oct 2026 · Python, FastAPI, PostgreSQL

Finds the root cause of small-satellite anomalies from a causal graph and incident memory. 93% root-cause accuracy vs 32% for a limit-check console; 70 s median detection.

LineCheck · AI aircraft inspector on FaceTime

Beta Fund × AWS × OpenAI hackathon · Oct 2026 · OpenAI Realtime, LiveKit

Technicians call it with hands on the aircraft: it inspects a photo of the part, cites FAA AC 43.13-1B, checks airworthiness directives and drafts the work order for a mechanic to sign.

Sepsis Early Warning · ICU prediction

Mar 2026 · XGBoost, Keras, Optuna

~0.91 AUROC seven hours before onset (Sepsis-3); 441 time-series features from 44 ICU signals, XGBoost tuned by Bayesian search.

PharmaSentinel · explainable clinical NLP

Mar 2026 · PyTorch, BERT, Docker

Multi-task BERT for drug sentiment, condition and rating across 215K+ reviews, with Integrated Gradients, uncertainty estimates and a fairness audit.

Hospital Readmission Predictor

Amazon ML Summer Program · scikit-learn, SHAP

30-day readmission risk for diabetic patients from 10 years of data across 130 US hospitals; stacking ensemble with SHAP explanations.

PROGRAMS

Google Cloud Gen AI Academy APAC, Code Vipassana and Startup School: multi-agent systems with ADK, Gemini, Vertex AI, AlloyDB and BigQuery · McKinsey Forward Program

HOW EVERY PROJECT WORKS

01 · SENSE	02 · MODEL	03 · VERIFY	04 · DECIDE
Read the signal	**Find the pattern**	**Prove it can be trusted**	**A human signs**
telemetry · vitals · images	XGBoost · LSTM · BERT	root causes · SHAP · citations	AI drafts, people decide

Scan for the full portfolio

harshitaadlakha.github.io
Research, six projects, live demos and replays.

THIS WEEK

22 Oct · Paris · FDIR Copilot
23 Oct · San Francisco · LineCheck
24 Oct · London · Canva Gen© Hackathon